1. OBJECTIVE

To determine the number of viable microorganisms in fish and fishery products.

2. SCOPE

This instruction is applicable in the determination of total bacterial counts found in fish, fishery and aquaculture products.

3. RESPONSIBILITY

The analyst performing the test should ensure that the procedure is followed accordingly.

4. APPARATUS/EQUIPMENT/CONSUMABLES

4.1 Alcohol / Fireboy

4.2 Disinfectant (70% solution isopropyl alcohol)

4.3 Autoclave

4.4 Autoclave tape

4.5 Top loading Balance

4.6 Aluminum Foil

4.7 Tissue paper or paper towel

4.8 Spatula

4.9 Forceps – stainless steel

4.10 Scissors – stainless steel

4.11 Can opener / knives

4.12 Graduated cylinder (100 ml and 500 ml, Pyrex)

4.13 Media bottles (500 ml and 1000 ml, Pyrex)

4.14 Dilution bottles with screw caps (160 ml, Pyrex)

4.15 Sterile Pipets (10 ml)

4.16 Sterile Pipettor

4.17 Sterile Pipet tips (100 – 1000 µl)

4.18 Sterile Petri dishes with cover (100x15mm, Pyrex)

4.19 Sterile stomacher bags

4.20 Stomacher

4.21 Incubator (maintained at 35 ± 1°C)

4.22 pH meter

4.23 Glass-marking pen

4.24 Biosafety Cabinet

4.25 Colony Counter

5. CULTURE MEDIA

5.1All information regarding media preparation is placed in LF-W04-01, Reagent Preparation Logbook.

5.2 Plate Count Agar

Suspend 22.5 g of the medium in one liter of distilled water. Mix thoroughly. Heat with frequent agitation and boil for 1 minute to completely dissolve the powder. Sterilize 15 min at 121 °C. Final pH, 7.0 ± 0.2. If agar solidifies, melt completely and temperature at 45 °C prior to use.

5.3 Butterfield's Phosphate-Buffered Dilution Water

Suspend 34 g of KH2PO4 to 500 liter of distilled water. Adjust pH to 7.2 with 1 N NaOH. Bring volume to 1 liter with distilled water. Sterilize 15 min at 121°C. Store in a refrigerator at 4°C.

Dilution Blanks

Take 1.25 ml of above stock solution and bring volume to 1 liter with distilled water. Dispense into bottles to 90 or 99 ± 1 ml. Sterilize 15 min at 121°C.

6. PROCEDURE

6.1 Weigh 25 g sample into a sterile stomacher bag.

6.2 Add 225 ml of Butterfield's phosphate-buffered dilution water to make 10^{-1} dilution. Homogenize using the stomacher for 2 m at 230 rpm. Using separate sterile pipettes, prepare decimal dilution of 10^{-2} and 10^{-3} of food homogenate by transferring 10 ml of the previous dilution to 90 ml diluents. Shake all dilutions immediately 25 times in 30 cm arc or vortex mix for 7s prior to transfer to ensure uniform distribution of the microorganisms present.

6.3 Using a pipettor with 100 – 1000 µl sterile pipette tips, pipet 1 ml of each dilution into each duplicate sterile Petri dishes (mark clearly the duplicate Petri dishes identifying sample, sample unit, dilution and date of inoculation). Use different pipette tips in each type of dilution. Discard used pipette tips into disinfectant jars.

6.4 Add 15-20 ml of melted and tempered Plate count agar (PCA) to each inoculated plates.

6.5 Mix the inoculum and PCA carefully and thoroughly by alternate rotation and back-and-forth motion of plates on flat level surface.

6.6 Allow the medium to solidify, invert the plates and incubate at 35˚C for 48 ± 2 h. Pour PCA to 2 sterile Petri dishes. Open one plate labelled as "air control" and cover after 15 minutes of exposure. Label unexposed plate as "medium control" for PCA.

6.7 Pipet 1 ml of sterile diluent into a Petri dish labelled as "diluent control" and pour agar.

6.8 After the required incubation period count the colonies and select those having 25-250 colonies.

6.9 Calculate the no. of organisms per gram. Express counts in colony-forming units per gram (CFU/g) of sample.

7. CALCULATIONS AND RECORDING COUNTS

To avoid creating a fictitious impression and accuracy when computing APC, report only the first two significant digits. Round off to two significant figures only at the time of conversion to SPC (Standard Plate Count). Round by raising the second digit to the next highest number when the third digit is 6, 7, 8 or 9 and use zero for each successive digit toward the right from the second digit. Round down when the third digit is 1, 2, 3 or 4. When the third digit is 5, round up when the second digit is odd and round down when the second digit is even.

Examples	
Calculated Count	APC
12,700	13,000
12,400	12,000
15,500	16,000
14,500	14,000

7.1 For plates with 25-250 CFU

Formula:

$$N = \frac{\Sigma C}{(1x\, n_1) + (0.1 \text{ x } n_2)}\, d$$

Calculate the APC as follows:

Where N = Number of colonies per ml or g of sample

ΣC = Sum of all colonies on all plates counted

n_1 = no. of plates with 25-250 colonies in lowest dilution

n_2 = no. of plates with 25-250 colonies in highest dilution

d = dilution from which the first counts were obtained

Example:

1:100	1:1000
232, 244	33, 28

$$N = \frac{(232+244+33+28)}{[(1x2)+(0.1x20)]10^{-2}}$$

$$= 24,409 \approx 24,000$$

When counts of duplicate plates fall within and without the 25-250 colony range, use only those counts that fall within this range.

7.2 For all plates with fewer than 25 CFU.

When plates from both dilutions yield fewer than 25 CFU each, record actual plate count but record the count (EAPC – Estimated Aerobic Plate Count) as less than 25 x 1/d when d is the dilution from which the first counts were obtained.

Example:

Colonies		
1:100	1:1000	EAPC/ml (g)
18	2	<2500
0	0	<100

7.3 For all plates with more than 250 CFU.

When plates from both 2 dilutions yield more than 250 CFU each (but fewer than $100/cm^2$), estimate the aerobic counts from the plates (EAPC) nearest 250 and multiply by the dilution.

Example:

Colonies		
1:100	1:1000	EAPC/ml (g)
TNTC (too numerous to count)	640	640,000

7.4 For all plates with spreaders and/or laboratory accident, report respectively as Spreader (SPR), or Laboratory Accident (LA).

7.5 For all plates with more than an average of 100 CFU per sq. cm. Estimate the APC as greater than 100 times the highest dilution plated, times the area of the plate.

Example:

Colonies / Dilution		
1:100	1:1000	EAPC/ml (g)
TNTC	7,150 [(a)]	<6,500,000 EAPC[(b)]
TNTC	6490	<5,900,000 EAPC

[a] Based on plate area of $65cm^2$ [b] EAPC, estimated APC

8. REFERENCES

- Bacteriological Analytical Manual, Edition 8, Revision A, 1998. Chapter 3. Authors: Larry Maturin and James T. Peeler
- British Standard Methods for Microbiological Examination of Food and Animal Stuffs, Part 5. 1981. Enumeration of Microorganisms – Colony Count at 30°C (Surface Plate Technique) BS 5763: Part 5.
- Food and Drug Administration. 1992. Bacteriological Analytical Manual, 7th ed. AOAC International, Arlington, VA, USA.

9. DOCUMENTS

LF W01-01, Reagent and Media Preparation Logbook
LF W01-MIC-03, Analyst Worksheet for Bacteriological Examination of Fish and Fishery Products